Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Full-Stack SaaS · Java & Spring Boot · Python · OOP & SOLID · PostgreSQL · REST APIs · AWS · Apache Spark

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Virginia, USA · open to Chicago

PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years designing, building, and owning full-stack SaaS modules in production — from architecting to deploying to maintaining. At Entrata, owned 20+ production modules end-to-end across the full stack (Java, Spring Boot, Python, PHP, JavaScript, PostgreSQL, RabbitMQ, AWS) for 100K+ monthly active users at 99.5% uptime. Deep expertise in object-oriented programming (OOP/SOLID/DRY), REST API design and SOA web services, PostgreSQL schema design and query optimization (200+ optimizations, 30% performance gain), and service-oriented architecture. Nice-to-haves covered: AWS Lambda + API Gateway + IAM, Apache Spark + PySpark + Scala, Data Warehouse design, Tableau and Power BI. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Designing and implementing ML models and full-stack data pipelines; building APIs and agentic AI components; debugging production reliability issues; rapid prototyping and iterating on solutions in a fast-moving startup environment.

Magizhs Tech Solution May 2025 – Aug 2025

AI/ML Engineer Intern · Remote

- Owned full product build end-to-end: designed and implemented a LangGraph-based agentic compliance system applying SOLID OOP principles throughout; REST API integration; RAG over regulatory corpus with dependency injection pattern for modular service composition.

George Mason University Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries ($R^2 = 0.977$); HuggingFace model fine-tuning + PyTorch inference API deployment; statistical modeling and validation. Python, SQL, and OOP workshops to 600+ students; provided one-on-one debugging support.

Entrata (PropTech / Real Estate SaaS) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Full-stack SaaS product ownership - architecting through maintaining** (Java, Spring Boot, Python, PHP, JavaScript, PostgreSQL, RabbitMQ, AWS, Kubernetes) — owned 20+ production modules end-to-end for 100K+ MAU at 99.5% uptime. Applied SOLID principles, DRY principles, and OOP design patterns throughout; implemented SOA web services and REST APIs across multi-tenant architecture; designed maintainable, high-performing, secure production code. Zero compliance violations, zero security incidents across 7 years.
- **Deep PostgreSQL / RDBMS expertise + REST API and OData design** — 200+ SQL query optimizations (30% performance gain via execution plan analysis, indexing, partitioning, stored procedure refactoring); schema migrations across hundreds of thousands of records with rollback preparation; REST and OData API design; Spring Boot IoC container and dependency injection; Hibernate-compatible ORM layer; RabbitMQ async processing with idempotency guarantees; CI/CD (GitHub Actions, Maven, Jenkins) with automated testing; AWS (S3, EC2, RDS, Lambda, Step Functions, API Gateway, IAM).
- **Technical leadership and mentoring** — mentored 5+ junior engineers; conducted 300+ peer code reviews; provided accurate effort estimates; collaborated with global engineering teams in India across 120+ Agile sprints; drove adoption of Python/Bash automation eliminating 80% of manual workload.

PROJECTS

Leasel: Full-Stack SaaS AI Product - REST APIs + OOP Architecture FASTAPI REST + SOA

Python · FastAPI · LangGraph · SQLite · ChromaDB · OpenAI API · LangSmith | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Full-stack AI product built with SOLID OOP design: FastAPI REST service layer, dependency injection for modular agent orchestration, 8-agent pipeline (Triage → Retrieval → Reasoning → Validation → Escalation), SQLite + ChromaDB data layer, and OpenAPI-documented endpoints. Owned full module lifecycle: architected, designed, developed, deployed, and evaluated. 93.6% precision, 87.9% F1 across 1,240 compliance test cases.

AgentProbe: Production-Grade Evaluation Service - FastAPI + CI/CD + Observability POSTGRESQL-COMPATIBLE CI/CD

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- Production-grade evaluation platform with 15-endpoint FastAPI REST service; OOP-architected evaluation engine (correctness, groundedness, latency, token efficiency, hallucination rate); CI/CD regression harness with golden datasets (GitHub Actions); multi-provider LLM routing with cost controls; OpenTelemetry + Prometheus + Grafana observability stack. Built to the same production quality standard expected of a SaaS module: maintainable, testable, monitored.

CrashClarity: Big Data Analysis - Apache Spark-compatible Pipeline (104K Records) LARGE-SCALE DATA DATA WAREHOUSE

Python · R · scikit-learn · Random Forest · Logistic Regression · Data Visualization | portfolio

- End-to-end ML pipeline on 104K+ records: feature engineering, Random Forest + Logistic Regression (85%+ accuracy), dimensional data modeling aligned with OLAP/data warehouse design patterns, deployed dashboard on GitHub Pages. Scalable pipeline architecture compatible with Apache Spark EMR processing patterns.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87 · Transcripts available on request

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Languages & OOP: Java (Spring Boot), Python (8+ yrs), Kotlin (JVM), Scala, TypeScript, JavaScript, PHP, SQL, PL/SQL, Bash — Object-Oriented Programming (OOP), SOLID Principles, DRY Principles, Dependency Injection, IoC Containers (Spring), ORM Frameworks, Hibernate-compatible ORM, Service-Oriented Architecture (SOA), REST Frameworks, Microservices, Design Patterns

Data & APIs: PostgreSQL (8+ yrs, query optimization, schema design, stored procedures, migrations), MySQL, Redis, RabbitMQ, REST API Design, OData APIs, OpenAPI, GraphQL, FastAPI, Django, Spring Boot, Angular, React.js, Node.js, Data Warehouse Design, Dimensional Modeling, OLAP Cubes, Semantic Layers, BI Systems, Tableau, Power BI, Apache Spark, PySpark, EMR-compatible, Apache Kafka, ETL, Data Modeling

Cloud, DevOps & Build: AWS (S3, EC2, RDS, Lambda, Step Functions, API Gateway, IAM), Docker, Kubernetes (K8s), Terraform, CI/CD (GitHub Actions, Jenkins), Maven, Build Automation, OpenTelemetry, Prometheus, Grafana, Multi-Tenant Architecture, PHI Compliance, HIPAA, SOC 2, Agile/Scrum (120+ sprints)